School of Design & Environment
Mechanical Facilities

Practical 1 Indoor Environmental Quality

Class Group		Practical Group: A/B/C/D/E	Date:
Reg. No.	Student No.	Full Name	Grades
			blank

Objective: To study the parameters of indoor environmental quality (IEQ).

Equipment Sling psychrometer, TSI VelociCalc, Sound Level and Lighting Level Meters

Instructions:

Submit the soft copy individually to BrightSpace. File Name Format:
Pract No._Class Group_Register No._Full name
e.g. Pract 1_1D2_1_full name

Indoor Environmental Quality covers the following areas in Singapore

The Singapore Standard SS 554: 2016 – Code of Practice for Indoor Air Quality for Air-Conditioned Buildings specifies indoor air quality as follow:

Physical Parameters	Acceptable Limit
Operative Temperature Relative Humidity Air Movement	**24 °C ± 1 °C** (23 °C – 25 °C) ≤ **65%** (for new buildings) ≤ 70% (for existing buildings) **0.2 ± 0.1 m/s**

Contaminant	Acceptable Limit
Carbon Dioxide, CO_2	**700 ppm > outdoor** (8h)

Green Mark NRD 2015

Area	**Sound Level Average dBA**
Open Offices	45
Classrooms	35

SS 531: 2013 - Code of Practice for Lighting of Work Places

Area/Activity	**Illuminance (Lux)**
General office - Normal	500

Procedure

1. Measure the parameters for the three locations as indicated in the table.

a) Use the sling psychrometer to measure the dry bulb and web temperatures, and determine the relative humidity from the online site.
b) Measure the dry bulb and wet bulb temperatures, relative humidity and air velocity using the TSI VelociCalc. Take the readings from the wall mounted sensors if available.
c) Take the carbon dioxide (CO_2) reading at the wall mounted sensors.
d) Measure sound and lighting level with meters provided and check with your phone app on sound level.

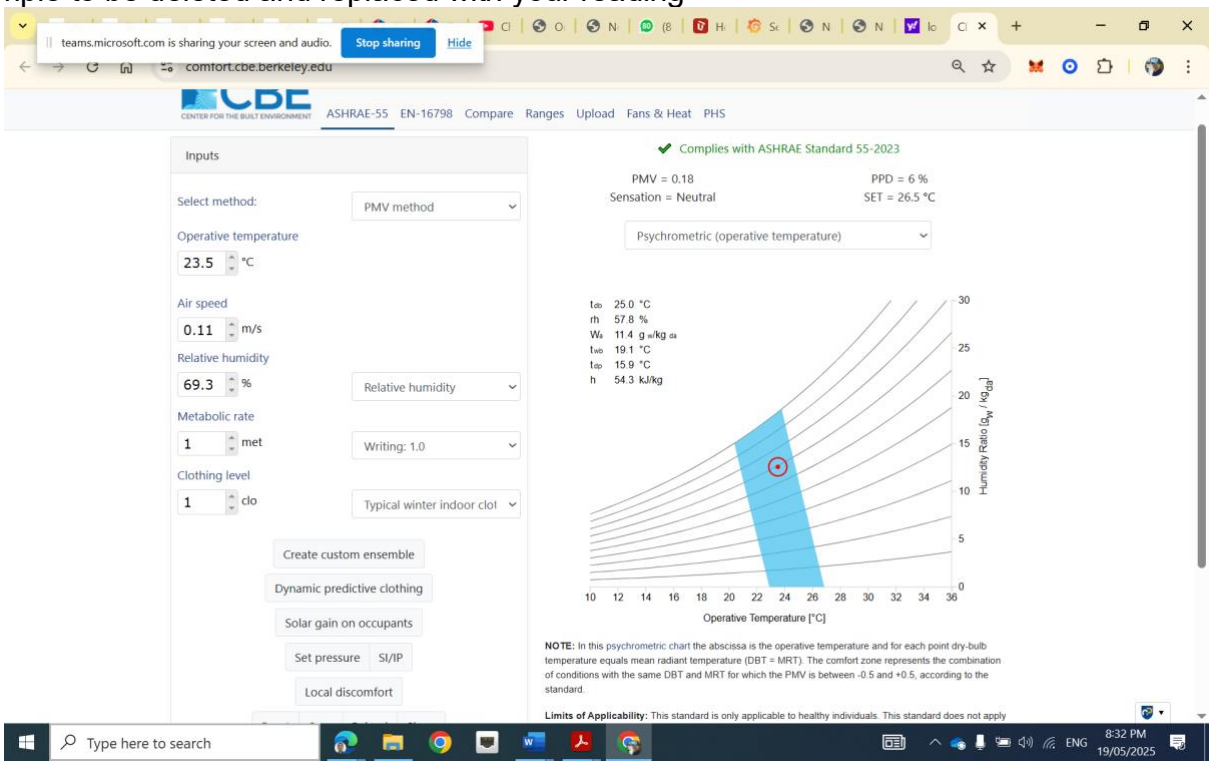
	Lab	Chiller Plant	Outdoor
Sling Psychrometer			
Dry Bulb Temperature, °C	23.5		27
Wet Bulb Temperature, °C	19.5		25
Relative Humidity*, %	69.3		
TSI VelociCalc			
Dry Bulb Temperature, °C	24.5		
Wet Bulb Temperature, °C	18.8		
Relative Humidity, %	58.9		
Air Velocity, m/s	0.08		0.53
Wall Mounted Sensor			
Carbon Dioxide, ppm			----
Sound Level Meter **Lux Meter**			
Sound level, dBA			
Lighting level, lux			

*Relative Humidity from sling psychrometer can be obtain from the following url:
https://daytonashrae.org/psychrometrics/psychrometrics_si.html#start

2. For the three locations,
    a. Go the online comfort study chart based on ASHRAE-55 standard http://comfort.cbe.berkeley.edu/
    b. For the three locations, input the temperature and relative humidity readings from the sling psychrometer, and air speed from the TSI VelociCalc, for other parameter use the same as in the sample
    c. Take screen shots of the reading and include them in this MS Word document.

a) Lab

Sample to be deleted and replaced with your reading



b) Chiller Plant Room

c) Outdoor

Discussion

1. State whether the three locations meet the indoor environmental quality requirements of the Singapore Standard and ASHRAE-55 thermal standard.

2. Compare the dry bulb temperature readings of the lab from
    a. Sling psychromter
    b. TSI VelociCalc

    Calculate the % difference for the lab.
    % difference = (A ~ B) / A x100%, A ~ B is the difference between A and B
    A is the temperature measured by the psychrometer

    (27 – 25)/25 x 100% = 8%

3. Which instrument would you would rely on? Explain your reasons.